

7-1 已知人的脑电波频率范围 0-45Hz, 对其作数字信号处理时, 求最大抽样周期 T 。如果 $T=5\text{ms}$ 抽样, 要使抽样信号通过一理想低通滤波器后, 能不失真的恢复原信号, 问理想低通滤波器的截止频率 f_c 应满足什么条件?

解: (1) 0.011s

(2) 45Hz~155Hz

7-2

某地址勘探测试设备给出的发射信号 $x(n) = \delta(n) + \frac{1}{2}\delta(n-1)$,

接收回波信号 $y(n) = \left(\frac{1}{2}\right)^n u(n)$, 若地层反射特性的系统函数用 $h(n)$ 表示, 且满足 $y(n) = h(n) * x(n)$ 。

求: $h(n)$;

解 (1)

$$h(0) = y(0)/x(0) = 1$$

$$h(1) = [y(1) - h(0)x(1)]/x(0) = \frac{1}{2} - \frac{1}{2} = 0$$

$$h(2) = \left[y(2) - \underbrace{h(0)x(2)}_{=0} - h(1)x(1) \right] / x(0) = \left(\frac{1}{2}\right)^2 - 0 = \left(\frac{1}{2}\right)^2$$

$$h(3) = \left[y(3) - \underbrace{h(0)x(3) - h(1)x(2) - h(2)x(1)}_{=0} \right] / x(0) = \left(\frac{1}{2}\right)^3 - \left(\frac{1}{2}\right)^2 \frac{1}{2} = 0$$

...

$$h(n) = \begin{cases} 0 & n \text{ 为奇数} \\ \left(\frac{1}{2}\right)^n & n \text{ 为偶数} \end{cases}$$

解 (2)

$$\begin{aligned} h(n) &= \left[y(n) - \sum_{m=0}^{n-1} h(m)x(n-m) \right] / x(0) \\ &= \left(\frac{1}{2}\right)^n u(n) - \sum_{m=0}^{n-1} h(m) \left[\delta(n-m) + \frac{1}{2}\delta(n-m-1) \right] \\ &= \left(\frac{1}{2}\right)^n u(n) - \frac{1}{2}h(n-1) \end{aligned}$$

$$h(n) + \frac{1}{2}h(n-1) = \left(\frac{1}{2}\right)^n u(n)$$

$$\frac{1}{2}h(n) + \frac{1}{2}\frac{1}{2}h(n-2) = \frac{1}{2}\left(\frac{1}{2}\right)^{n-1} u(n-1)$$

7-4: 测 2-17

$$f_s = 2f_m = 2 \times 8.3 = 16.6 \text{ kHz}$$

$$N = \frac{f_s}{\Delta f} = \frac{16600}{1} = 1.66 \times 10^4$$